Forced weakly dumped coupled oscillator experiment

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A forced weakly dumped harmonic oscillator linear model having one and two degrees of freedom is considered. In Laboratory 3 it is implemented on air-track experiment set. The model expectations for amplitude and phase response are compared with measurements. It is shown that measurements agree well enough with the model.

One oscillator case

One degree of freedom weakly dumped oscillator cart having coordinate $x(t)$ and mass m is connected on the right side to spring with elastic constant k_2, and on the left side by spring with elastic constant k_1 to harmonic driver cart with coordinate $\eta(t) = \eta_0 \cos \omega t$ moving with frequency ω and amplitude η_0. (see figure 1).

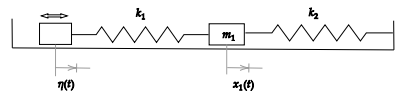

Figure 1: One degree of freedom forced oscillator set with parameters: $m = 209$ g, $k_1 = k_2 = 4.48$ H/m, mass of each spring $m_k = 17$ g. Quality factor $Q = 24$ (magnets are applied vertically on the cart, creating maximal viscous friction). Natural frequency $f = 1.00$ Hz.

Let in this section denote oscillator elasticity $k \equiv k_1 + k_2$.

Motion equation is

$$m\ddot{x} = -k_1(x - \eta) - k_2 x - b\dot{x}, \tag{1}$$

A note on effective masses. Effective oscillator mass m is a sum of cart mass and part of adjacent springs masses: $m = m_{\texttt{cart}} + \beta \cdot m_{\texttt{springs}}$. Theoretical value $\beta = 1/3$. True value can be found experimentally.

where b is viscous friction coefficient, expressed by quality factor Q:

$$b = \frac{\sqrt{km}}{Q} \tag{2}$$

Quality factor Q can be found by measuring oscillator amplitude decay in free oscillations state.

Equation 1 rewritten in complex form will be

$$m\ddot{x} + b\dot{x} + kx = \eta_0 k_1 e^{i\omega t} \tag{3}$$

Let's look for steady-state solution in complex form:

$$x(t) = ce^{i\omega t}. \tag{4}$$

Here c is complex amplitude with real part called *elastic amplitude* and imaginary part called *absorptive amplitude* (see [1], p.105).

[1] Frank S Crawford. *Waves.* Berkeley physics course vol. 3. McGraw-Hill, New York, 1968. ISBN 0070048606

Substitute solution 4 into motion equation 3:

$$ce^{i\omega t}(-m\omega^2 + ib\omega + km) = \eta_0 k_1 e^{i\omega t} \tag{5}$$

Therefore relative amplitude response c/η_0 is

$$\frac{c}{\eta_0} = \frac{k_1}{-m\omega^2 + ib\omega + km} \tag{6}$$

Experiment results are shown on Figure 2.

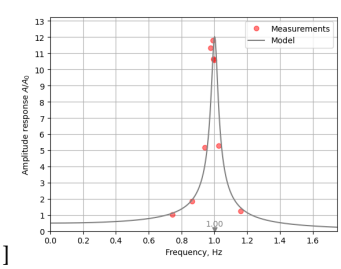

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Figure 2: Amplitude response experiment with experimental points. Parameters as on fig 1. Natural frequency f=1.00 Hz is shown by triangle marker on frequencies axis.

Coupled scillator case

A two-degree forced oscillator as shown on Figure 3.

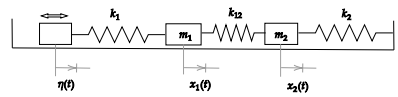


Figure 3: Coupled oscillator. Cart 2 with mass m_2 is coupled with wall by string with elastic coefficient k_2 and to the Cart 1 with mass m_1 by spring k_{12}. On the left end Cart 1 is coupled with driver by spring with elastic coefficient k_1. Driver is moving with amplitude η_0 and frequency ω. Parameters: $m_1 = 209$ g, $m_2 = 214$ g, $k_1 = k_2 = k_{12} = 4.48$ H/m, mass of each spring $m_k = 17$ g. Magnets set vertically. $Q_1 = 26$, $Q_2 = 25$.

Model equations

$$\begin{aligned} m_1\ddot{x}_1 + b_1\dot{x}_1 + (k_1 + k_{12})x_1 - k_{12}x_2 &= \eta_0 k_1 e^{i\omega t}, \\ m_2\ddot{x}_2 + b_2\dot{x}_2 + (k_2 + k_{12})x_2 - k_{12}x_1 &= 0 \end{aligned} \tag{7}$$

The same in matrix form: Coordinates vector

$$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \tag{8}$$

Masses matrix

$$M = \begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix}, \tag{9}$$

Elastic coefficients matrix

$$K = \begin{bmatrix} k_1 + k_{12} & -k_{12} \\ -k_{12} & k_{12} + k_2 \end{bmatrix}, \tag{10}$$

Friction matrix

$$B = \begin{bmatrix} b_1 & 0 \\ 0 & b_2 \end{bmatrix}, \tag{11}$$

Force amplitude vector

$$F = \eta_0 \begin{bmatrix} k_1 \\ 0 \end{bmatrix}, \tag{12}$$

Equation for forced oscillations with friction

$$M\ddot{X}(t) + B\dot{X}(t) + KX(t) = Fe^{i\omega t} \tag{13}$$

Natural frequencies and normal coordinates

Equation of free oscillations without friction

$$M\ddot{X}(t) + KX(t) = 0 \tag{14}$$

Let I - diagonal identity matrix, and $Z = M^{-1}K$ - mechanical impedance matrix. Then eigenvalues λ_i and eigenvectors matrix E are found from equation

$$\det|Z - \lambda I| = \begin{vmatrix} \frac{k_1+k_{12}}{m_1} - \lambda & -\frac{k_{12}}{m_1} \\ -\frac{k_{12}}{m_2} & \frac{k_{12}+k_2}{m_2} - \lambda \end{vmatrix} = 0 \tag{15}$$

Eigenvectors are squares of the oscillator natural frequencies:

$$\omega_i^2 = \lambda_i. \tag{16}$$

For this experimental system natural frequencies converted from radians per second to cycles per second are 1.22 Hz and 0.71 Hz.

Conversion from original to normal coordinates and vise versa is linear transformation by eigenmatrix:

$$x = Ey, \; y = E^{-1}x \tag{17}$$

In normal coordinates non-dumped oscillations are decoupled for any initial conditions.

Amplitude and phase response

Model equations for steady-state solution are:

$$X(t) = Ce^{i\omega t}, \tag{18}$$

where C is vector of complex amplitudes:

$$C = \begin{bmatrix} C_1 \\ C_2 \end{bmatrix}. \tag{19}$$

Substitute steady-state solution to motion equation to find C:

$$(-\omega^2 M + i\omega B + K) \times C = F \tag{20}$$

Let

$$Z = -\omega^2 M + i\omega B + K. \tag{21}$$

Relative amplitudes response

$$\frac{C}{\eta_0} = Z^{-1}\frac{F}{\eta_0} \tag{22}$$

Experiment results are shown on figure 5. Friction constants b_i and quality factors Q_i are found by measuring free oscillations decay for each mass and adjacent springs separately. A maximum relative displacement is expected from the model response with known quality. Note that system with two degrees of freedom has lower maximal response at resonance than the system with one oscillator of the same quality. Note also a "still point" — a point on driver frequency axis between low and high resonance frequencies, where amplitude C_1 becomes nearly zero.

The results of experiment agree well with the model.

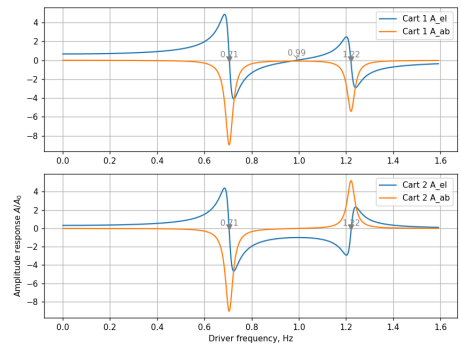
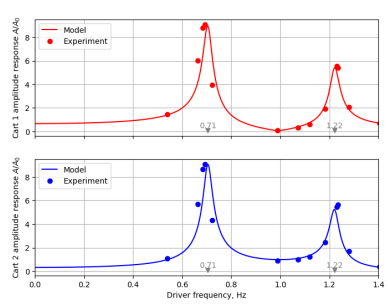
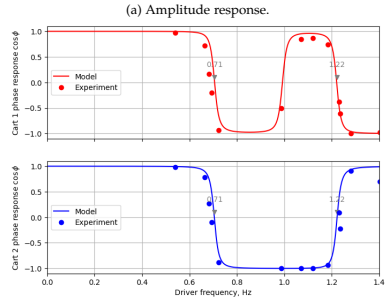


Figure 4: Expected amplitude response. The negative sign of amplitude means that the phase is opposite to driver displacement phase. Note the still point of Cart 1 between resonances: at this frequency (0.99Hz) amplitude of mass 1 becomes nearly zero, while mass 2 oscillates with the amplitude of driver.



(a) Amplitude response.



(b) Phase response

Figure 5: Amplitude and phase response of coupled oscillator. Lines are theoretical response amplitudes, points are experimental data. Parameters: $m_1 = 209$ g, $m_2 = 214$ g, $k_1 = k_2 = k_{12} = 4.48$ H/m, mass of each spring $m_k = 17$ g. Magnets set vertically. $Q_1 = 26$, $Q_2 = 25$.

References

Frank S Crawford. *Waves.* Berkeley physics course vol. 3. McGraw-Hill, New York, 1968. ISBN 0070048606.

Appendix. Python code snippets used for data analysis

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import linregress
from scipy.signal import find_peaks, peak_prominences
```

Listing 1: Modules import

Full code in Jupyter Notebook see in GitHub repository `https://github.com/tphlabs/Lab3_CO_engineer`

```
def get_signal(url, counts=238, length=0.125):
  data = np.loadtxt(url).T
  t = data[1, :]
  x = data[2:, :] /counts * length # convert from counts to meters
  # how many signals: Kruze may return  from 1 up to 4
  signals = x.shape[0]
  # shift mean to zero
  for i in range(signals):
    x[i] -= x[i].mean()
  # returns t and signals x1, x2, ..
  return tuple([t] + [x[i] for i in range(signals)])
```

Listing 2: Get signal from Kruze interface

```
def get_peaks(t, signal):
  rate = np.median(np.diff(t))
  min_period = 0.2 #s
  min_prominence = 2e-3 ## 2 mm
  peaks, _ = find_peaks(signal, prominence=min_prominence,
                        distance=min_period/rate )
  return t[peaks], signal[peaks]
```

Listing 3: Get signal peaks

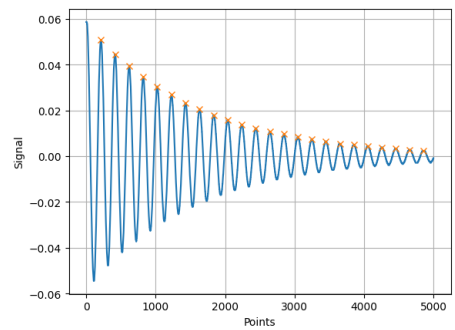


Figure 6: Getting peaks from a signal is a key to calculate harmonic signal amplitudes, find frequency and estimate uncertainty.

```
def get_quality(t, harmonic_signal):
  tpeaks, peaks = get_peaks(t, harmonic_signal)
  decay = np.log(peaks / peaks[0])
  cycles = range(len(peaks))
  fit = linregress(cycles, decay)
  quality = -np.pi/fit.slope
  return quality
```

Listing 4: Get harmonic oscillator quality

```
def get_frequency(t, harmonic_signal):
  tpeaks, _ = get_peaks(t, harmonic_signal)
  periods = np.diff(tpeaks)
  f0 = 1 / periods
  return np.mean(f0), np.std(f0)
```

Listing 5: Get signal frequency

```
def get_cosine(x1, x2):
# x1, x2 - vectors of harmonic signal x(t)
# Operator @ stands for scalar multiplication ov vectors
  x11 = x1 @ x1
  x12 = x1 @ x2
  x22 = x2 @ x2
  cosine = x12 / np.sqrt(x11 * x22)
  return cosine
```

Listing 6: Get cosine of phase difference between two harmonic signals, having the same frequency